layouts and JavaScript variables, thus demonstrating the power of Firebug. As can be seen in the same screen shot, the CSS section on the right also shows the line numbers of the .css file, which produces that page section. This really helps developers gain complete visibility of what needs to be modified in their code.

Firebug is both an inspector and an editor. All objects, i.e., HTML, CSS and JavaScript can be edited with a single or double click. When things go wrong, Firebug lets you know immediately, and gives you detailed and useful information about errors in JavaScript, CSS and XML. As you make changes in Firebug, you can see their result or effect in the browser instantly. All changes made in Firebug are in real time, so as you refresh your browser, it will show you the page before the changes were made. Firebug just helps you to get a particular CSS file with the line number or particular HTML tag on a page to make changes from your local machine, and then you can make permanent changes the way you want. It allows you to add new properties in CSS.

Now let's look at a few advanced features of Firebug. As you can see on the panel, it has multiple views to drill down into the code, based on which code area needs to be targeted for debugging. There are two HTML panes, i.e., the Node view pane and the HTML side panels.

- **The Node view**, which is on the left, allows you to inspect and modify the HTML tags of the page. When you hover the mouse over the HTML tags, it shows a section being highlighted on the page, which points to the location of elements placed on the page.
- **HTML side panels** are on the right, and give more information about the styles of the highlighted elements.

There are four side panels: style, computed, layout and DOM.

Style: It shows you which CSS lines affect what part of your code. It also shows which style sheet effects correspond to that part, along with the line numbers.

Computed: It shows the CSS with properties such as font size, type, alignment, etc. This gives information about how HTML renders that element.

Layout: It shows you the graphical box model of the element that you have clicked. It helps you to modify the values of the padding, margin and border by double clicking on it in real time.

DOM: It shows the document object model of the selected element. It's useful for advanced client-side script debugging.

Console view

The console is for logging, tracing, profiling and the command line. The console shows JavaScript error message logs. It can also be used to run JavaScript commands.

CSS view

The main difference between styles in the HTML panel and CSS view is that it allows you to work on all CSS styles, and not just the highlighted one.

Script view

It allows you to jump to the particular section of the JavaScript that you have selected. It allows debugging of functions, shows the stack of functions in real time, the list of currently active breakpoints, etc. It includes JavaScript Debugger, which allows you to pause at any time. This feature is important to debug code snippets that are sluggish and need performance tuning.

DOM view

The Document Object Model is a great big hierarchy of objects and functions. Firebug helps you find DOM objects quickly and then edit them on the fly.

Net view

It gives information about how long it will take for your page to load or which request takes more time to load. The slowness of page loading can be attributed to static code, references to other pages, heavy images, etc. Net view helps you to pin point the exact section of a file which is slowing down the performance. It also gives suggestions regarding which images should be compressed for better performance, etc.

Cookies view

The cookies panel shows you the cookies on your current or selected page. You can deny cookies for a particular site, create cookies and also delete cookies, on the fly. Besides these features, there is a search box available on the panel which can be used to find elements or the sections of code being debugged. The panel dashboard can be detached from the browser, and be placed on screen as a floating entity for ease of use.

While Firebug is a self-contained environment for debugging, there are a few limitations which will hopefully be fixed in future versions. First, the CSS panel doesn't allow the edit option. Second, while debugging HTML elements with hover properties, it is cumbersome to use Firebug to highlight a particular section and debug it. This is true in case of client-side slide show containers, which may or may not be using JQuery. END

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